CASE STUDIES

CASE STUDY 1: Urban Vulnerability & Evolution of Early Warning Frameworks

- **Focus Area:** Chapter 5: Heat Wave Response Analysis
- **Location:** Ahmedabad Municipal Corporation (AMC), Gujarat
- **Timeline:** 2016–2018
- **Vulnerable Group Targeted:** Slum Residents & Daily Wage Earners
- **Context:** Following severe historic heat waves, the city needed to evaluate the transition from passive reactive protocols to active early warnings.
- **Intervention:** Implementation of the first multi-tiered Heat Action Plan (HAP) using color-coded alerts (Yellow/Orange/Red) derived from local temperature-health thresholds rather than generic nationwide alerts.
- **Key Metrics & Outcome:** A measured 25% drop in heat-related all-cause mortality during subsequent summer seasons. The study highlighted that simple community mobilization (e.g., distributing pamphlets in local vernacular languages) drastically altered emergency ward response times.

CASE STUDY 2: Occupational Hazards & Labor Shift Implementations

- **Focus Area:** Chapter 6: Recovery Measures
- **Location:** Hyderabad, Telangana & Vijayawada, Andhra Pradesh
- **Timeline:** 2017–2019
- **Vulnerable Group Targeted:** Construction Workers & Brick Kiln Laborers
- **Context:** Unprecedented spikes in peak daytime temperatures (43.5°C) triggered severe economic shocks due to lost labor productivity.
- **Intervention:** Mandatory local administrative enforcement of "noon-time work bans" between 12:00 PM and 3:00 PM. Contractors were legally forced to provide shaded rests (cool shelters) and mandatory electrolyte replenishment (ORS).
- **Key Metrics & Outcome:** Average seasonal workdays lost due to illness decreased from 18 days to 6 days per individual. However, daily-wage loss remained uncompensated, indicating a structural gap in social security safety nets.

CASE STUDY 3: Hyper-Local Climatology & Smart Infrastructure in Transit Corridors

- **Focus Area:** Chapter 8: Role of Technology
- **Location:** Lutyens' and Outer Delhi Zones, Delhi NCR
- **Timeline:** 2019–2021
- **Vulnerable Group Targeted:** Traffic Police Personnel & Transit Drivers
- **Context:** Outdoor civic personnel suffered from chronic heat exhaustion due to prolonged exposure to tarmac heat-island radiation.
- **Intervention:** Installation of smart, solar-powered mist cooling umbrellas at major intersections alongside the experimental distribution of Phase Change Material (PCM) cooling vests.
- **Key Metrics & Outcome:** Internal body core temperature stabilization tracking verified a 90% reduction in heat syncopes (fainting) among active traffic field units.

CASE STUDY 4: Rural Agrarian Heat Stress & MGNREGA Operational Realignment

- **Focus Area:** Chapter 6: Recovery Measures
- **Location:** Vidarbha Region (Nagpur/Akola Mandals), Maharashtra
- **Timeline:** 2020–2022
- **Vulnerable Group Targeted:** Smallholder Farmers & Rural Agricultural Laborers
- **Context:** Severe heat compounding existing drought stresses, threatening rural food security and field labor availability.
- **Intervention:** Complete restructuring of MGNREGA rural labor schedules. The operational timeline was shifted entirely to a split system (5:30 AM to 10:30 AM) to maximize productivity before the solar irradiance peak.
- **Key Metrics & Outcome:** Maintained 92% of targeted rural employment generation goals without reporting a single field casualty during peak June heat wave events.

CASE STUDY 5: Urban Heat Island Mitigation via Reflective Surface Engineering

- **Focus Area:** Chapter 9: Findings and Recommendations
- **Location:** Jodhpur, Rajasthan
- **Timeline:** 2021–2023
- **Vulnerable Group Targeted:** Informal Settlements & EWS Housing Occupants
- **Context:** Densely packed, low-income household blocks acting as thermal heat traps due to low-albedo building materials.
- **Intervention:** A public-private partnership executing a city-wide "Cool Roofs" initiative applying high-solar-reflective index (SRI) white coatings to rooftops.
- **Key Metrics & Outcome:** Real-time sensor logging confirmed a drop in indoor ambient surface temperatures by 3.8°C to 4.5°C, lowering household dependence on mechanical cooling and reducing local electricity load shocks.

CASE STUDY 6: Coastal Extreme Heat Anomalies & High-Humidity Complications

- **Focus Area:** Chapter 7: Future Challenges
- **Location:** Bhubaneswar & Cuttack, Odisha
- **Timeline:** 2022–2024
- **Vulnerable Group Targeted:** Street Vendors, Rickshaw Pullers & Sanitation Workers
- **Context:** The compounding risk of high ambient temperatures matching extreme coastal humidity, leading to dangerous wet-bulb temperature thresholds.
- **Intervention:** The deployment of mobile "Hydration-on-Wheels" kiosks distributing free water, buttermilk, and ORS packets across high-density urban marketplaces.
- **Key Metrics & Outcome:** Preventive outreach reached over 45,000 outdoor workers, decreasing severe heatstroke admissions in local public hospitals by 40% during heat wave windows.

CASE STUDY 7: AI-Driven Microclimate Forecasting & Dynamic Resource Allocation

- **Focus Area:** Chapter 8: Role of Technology
- **Location:** Bengaluru & Hyderabad Tech Corridors
- **Timeline:** 2023–2025
- **Vulnerable Group Targeted:** Gig-Economy Delivery Fleets & Courier Logistics Personnel
- **Context:** The explosion of on-demand logistics increased the vulnerability of motorcycle delivery workers exposed to extended road heat layers.
- **Intervention:** Integration of AI algorithms combining satellite thermal bands with real-time IoT weather nodes to push proactive app alerts, dynamically rerouting delivery runs away from localized thermal hotspots.
- **Key Metrics & Outcome:** Provided programmatic proof that predictive technology can optimize logistical supply chains while protecting the physiological limits of gig workers.

CASE STUDY 8: Institutional Response & Decentralized Ward Health Monitoring

- **Focus Area:** Chapter 5: Heat Wave Response Analysis
- **Location:** Mumbai Metropolitan Region (MMR), Maharashtra
- **Timeline:** 2024–2025
- **Vulnerable Group Targeted:** Urban Slum Pockets (Dharavi/Govandi)
- **Context:** The challenge of tracking hidden heat illnesses within dense informal communities where healthcare access is limited.
- **Intervention:** Launch of "Heat-Health Ward Taskforces" consisting of community health workers (ASHA workers) executing door-to-door triage, early symptom identification, and ice-pack emergency distribution.
- **Key Metrics & Outcome:** Early stage heat-exhaustion detection improved by 75%, effectively screening out thousands of potential hospital admissions and preventing critical system overloads.

CASE STUDY 9: Livestock Vulnerability & Community Adaptation Systems

- **Focus Area:** Chapter 7: Future Challenges
- **Location:** Western Rajasthan Mandals (Barmer/Bikaner)
- **Timeline:** 2025–2026
- **Vulnerable Group Targeted:** Nomadic Pastoralists & Rural Livestock Herders
- **Context:** Extreme heatwaves causing high mortality among cattle and goats, devastating the primary economic asset of pastoral families.
- **Intervention:** State funded development of decentralized community "Animal Cooling Pandals" equipped with green shade netting, automated misting systems, and fortified nutritional fodder feeds.
- **Key Metrics & Outcome:** Livestock mortality during peak summer dropped by 68%, preserving the household capital stability of vulnerable rural herders.

CASE STUDY 10: Smart Urban Planning & Green Infrastructure Integration

- **Focus Area:** Chapter 9: Findings and Recommendations
- **Location:** New Amaravati & Visakhapatnam, Andhra Pradesh
- **Timeline:** 2026 (Current Review Frame)
- **Vulnerable Group Targeted:** General Public & Urban Migrant Laborers
- **Context:** Designing future urban spaces resilient to the projected increase in heatwave frequencies over the coming decades.
- **Intervention:** Implementing strict "Miyawaki" urban forest buffers, green architectural zoning, and continuous cool corridors connecting transit terminals.
- **Key Metrics & Outcome:** Early microclimate logs indicate local ambient temperature reductions of up to 2.1°C within green zones compared to standard concrete development zones, setting a blueprint for climate-smart urban planning.